

Case Study 2: Hospital Patient Care Analytics Pipeline

Scenario

A multi-specialty hospital wants to improve patient care, reduce waiting times, and predict high-risk patients. The hospital collects data from multiple systems such as patient registration, appointment scheduling, laboratory reports, wearable health devices, and doctor consultation notes.

The management wants to build a data-driven solution that can collect, process, store, and analyze patient data to improve operational efficiency and patient outcomes.

Your Task

Using the concepts learned in Data Engineering Fundamentals, design an end-to-end data engineering solution for the hospital.

Our task is to design a data engineering system that takes hospital data from different sources and turns it into useful information.

1. Data Sources

Data sources are the places from where the hospital gets patient data.

For this hospital, the main data sources are:

1. Patient Registration → Patient details
2. Appointment System → Appointment and doctor details
3. Laboratory Reports → Test results
4. Wearable Devices → Heart rate, blood pressure, oxygen level, etc.
5. Doctor Consultation Notes → Symptoms, diagnosis, treatment, medicines

2. Data Ingestion

Batch Ingestion

Data is collected and processed at scheduled times.

Examples in hospital:

- Appointment data → processed daily
- Laboratory reports → processed periodically
- Daily patient records → processed in batches

Real-Time / Stream Ingestion

Data is collected and processed immediately as it is generated.

Examples:

- Wearable devices → heart rate, oxygen level, blood pressure
- Emergency patient monitoring
- Real-time vital signs

3. Data Integration

Data Integration means combining and connecting data from different hospital systems so that we can get a complete view of a patient.

Hospital Example

Different systems have different information:

- Patient Registration → Patient details
- Appointment System → Appointment details
- Laboratory → Test results
- Wearable Devices → Health readings
- Doctor Notes → Diagnosis and treatment

We connect these using a common Patient_ID.

4. ETL / Data Transformation

ETL stands for Extract, Transform, and Load. It is the process of collecting data from different hospital sources, cleaning and preparing the data, and storing it for further analysis.

Extract

Data is collected from different hospital sources such as:

- Patient Registration
- Appointment System
- Laboratory Reports
- Wearable Health Devices
- Doctor Consultation Notes

Transform

The collected data is cleaned and prepared for analysis. This includes:

- Removing duplicate records

- Handling missing values
- Correcting invalid data
- Standardizing date and time formats
- Converting measurement units into a common format
- Connecting patient information using Patient_ID

Load

After transformation, the cleaned and structured data is loaded into a **Data Warehouse** for reporting and analysis. Raw data can also be retained in a **Data Lake** for future processing and AI applications.

5. Data Storage

Data Storage is the process of storing hospital data safely so that it can be accessed, processed, and analyzed when required.

For this hospital case, we can use three types of storage:

1. Database

Used for structured and operational data.

Examples:

- Patient details
- Appointment details
- Doctor details
- Inventory/administrative records

2. Data Lake

Used to store large amounts of raw data in different formats.

Examples:

- Wearable device data
- Raw laboratory files
- Doctor consultation notes
- Application/system logs

3. Data Warehouse

Used to store cleaned and structured data for reporting and analysis.

Examples:

- Patient information
- Appointment history
- Laboratory results
- Patient waiting times
- Processed health measurements

6. Data Modeling

Data modeling means organizing data into different tables and connecting related data.

In our hospital case:

- Patient Table → Patient details
- Appointment Table → Appointment details
- Lab Table → Test results
- Wearable Table → Heart rate, oxygen level, etc.
- Consultation Table → Diagnosis and treatment

These tables can be connected using Patient_ID.

Simple Example

Patient P101

- Patient details
- Appointment details
- Lab results
- Wearable readings
- Doctor consultation

7. Data Quality & Validation

Data Quality & Validation means checking whether the hospital data is correct, complete, and reliable before using it for analysis.

In our hospital case, we check for:

- Missing data → Patient age or Patient_ID is missing
- Duplicate data → Same patient record appears twice
- Incorrect data → Age = 200
- Invalid values → Heart rate or test result is outside a valid range
- Wrong format → Different date formats

- Inconsistent data → Same patient has different information in different systems.

8. Data Pipelines

A data pipeline is an automated flow that collects data from different hospital sources, processes it, and moves it to the required storage or analytics system.

Hospital Data Pipeline Flow

Hospital Data Sources

(Patient Registration, Appointment System, Laboratory Reports, Wearable Devices, Doctor Consultation Notes)

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Data Ingestion

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Data Integration

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ETL / Data Transformation

(Clean and Prepare Data)

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Data Quality & Validation

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Data Storage

(Data Lake / Data Warehouse)

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Analytics & AI Applications

9. Analytics & AI Applications

Analytics and AI applications use the processed hospital data to understand patient conditions, improve hospital operations, and support better decision-making.

Applications in Our Hospital Case

1. Patient Waiting Time Analysis
Analyze appointment and hospital data to identify long waiting times and improve scheduling.
2. High-Risk Patient Prediction
Use patient history, laboratory results, and wearable data to identify patients who may be at higher risk.
3. Patient Monitoring
Analyze real-time wearable data such as heart rate, blood pressure, and oxygen levels.
4. Appointment Analysis
Analyze appointment data to identify delays, cancellations, and busy time periods.

5. Hospital Performance Analysis

Use dashboards and reports to understand patient flow, doctor workload, and hospital efficiency.

6. Early Risk Detection

AI models can identify unusual patterns in patient data and alert healthcare professionals for further attention.

Simple Flow

Processed Hospital Data → Analytics / AI → Insights → Better Decisions → Improved Patient Care

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